MIL-S-36469A
26 August 1966

Superseding
MIL-S-36469
11 February 1966

MILITARY SPECIFICATION

SOLDERING FRAME AND TRIPOD, LABORATORY APPARATUS, DENTAL

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers soldering frames and tripods suitable for dental laboratory use.

1.2 Classification.

1.2.1 Types. Soldering frame and tripod shall be furnished in the following types, as specified: (see 6.1).

Type I - Soldering frame. Woven wire, 4-1/2 inches square.

Type II - Tripod. Spider grate type, 4-1/2 inches in diameter, 5 inches high.

	6520
FSC	6640

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:39Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical, General Specifications.
QQ-I-666	Iron Castings, Malleable.
QQ-N-281	Nickel-Copper-Alloy "Monel and R-Monel" Bars, Plates, Rods, Sheets, Strips, Wire Forgings, And Structural And Special Shaped Sections.
VV-L-800	Lubricating Oil, General Purpose, Preservative, "Water Displacing, Low Temperature."
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:39Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I (Soldering frame). The soldering frame shall be fabricated of alloy wire conforming to one of the following specifications, for chemical composition only:

(a) Nickel-copper alloy: Class A of QQ-N-281.
(b) Corrosion-resisting steel: Class 3 of GG-I-526.

3.1.2 Type II (Tripod).

3.1.2.1 Spider grate. The spider grate shall be cast of a good grade of iron, heat treated to produce a thoroughly malleable, tough casting. The casting shall meet the requirements for lug or fracture test only of QQ-I-666 or the distortion test specified herein.

3.1.2.2 Legs. The legs shall be made of a commercial grade of steel strip.

3.1.2.3 Bolts and nuts. The bolts and nuts shall be steel.

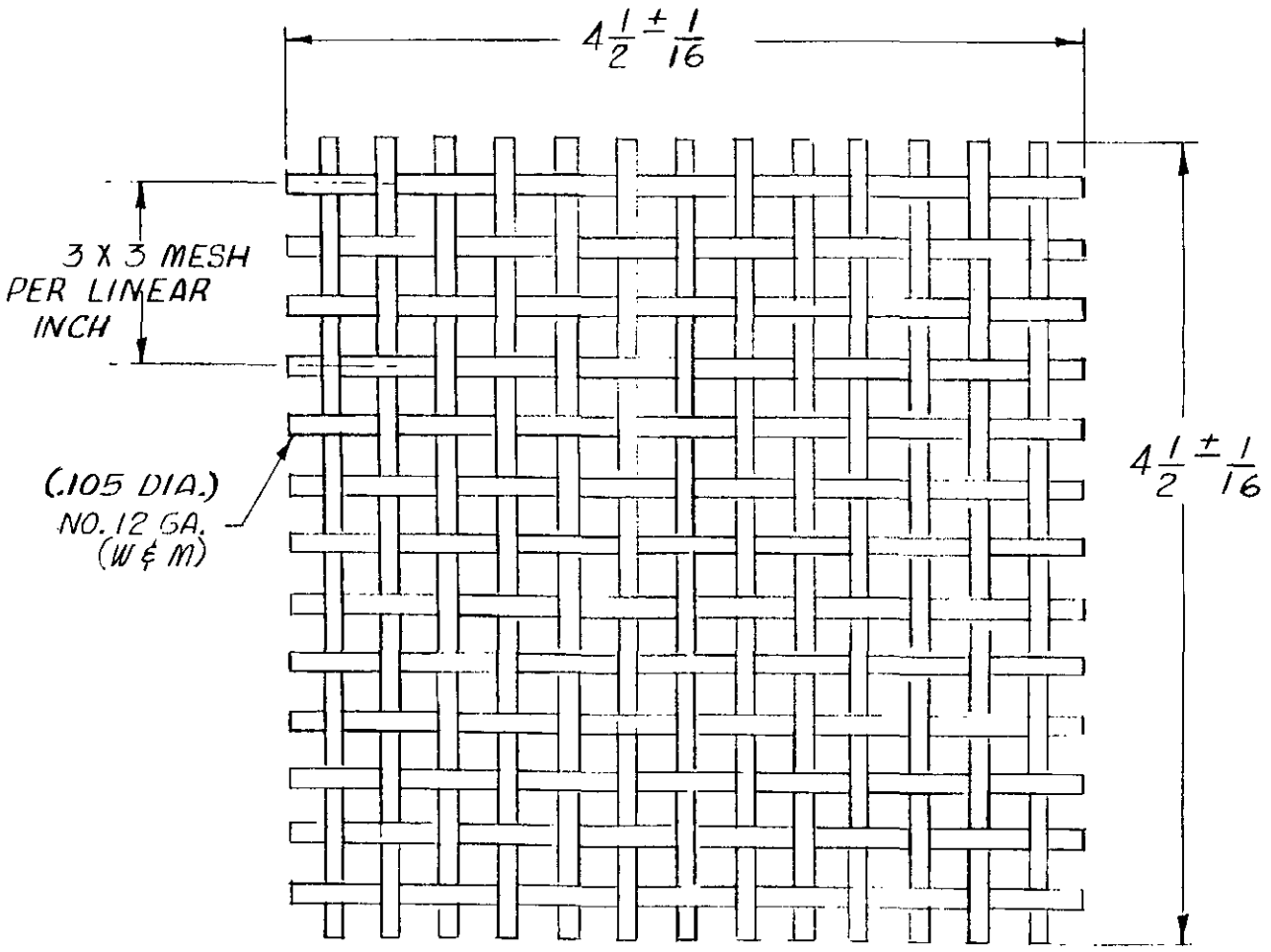
3.2 Design and dimensions.

3.2.1 Type I (Soldering frame). The soldering frame shall be of the design and dimensions shown in Figure 1. The wire shall be 0.105 ± 0.005 inch in diameter, and shall be wooven three wires to the inch, in both directions. The soldering frame shall be flat, or without noticeable curvature, when placed on a level surface.

3.2.2 Type II (Tripod).

3.2.2.1 Spider grate. The spider grate shall be of the design and dimensions shown in Figure 2. The center collar ring and all radial grate ribs shall provide a flat working surface, in order to provide uniform support for a flat work-load. The grate shall be flat, or without noticeable rock, when placed on a level surface. The leg seat areas shall be flat so that the legs rest properly, without rocking.

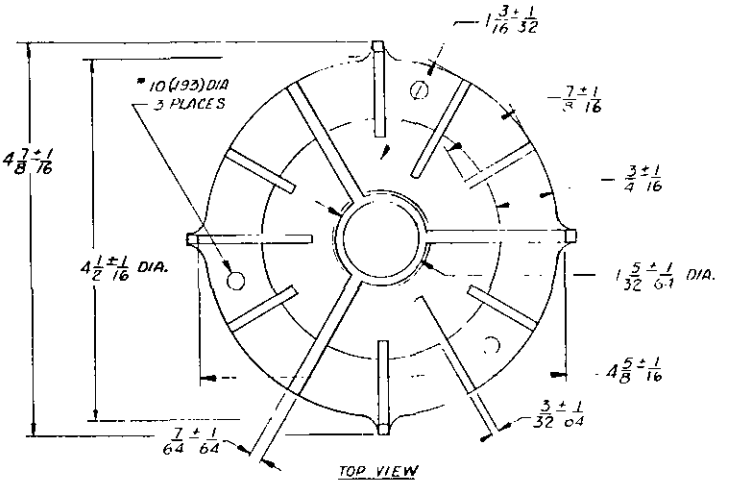
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:39Z
Check the source to verify that this is the current version before use.



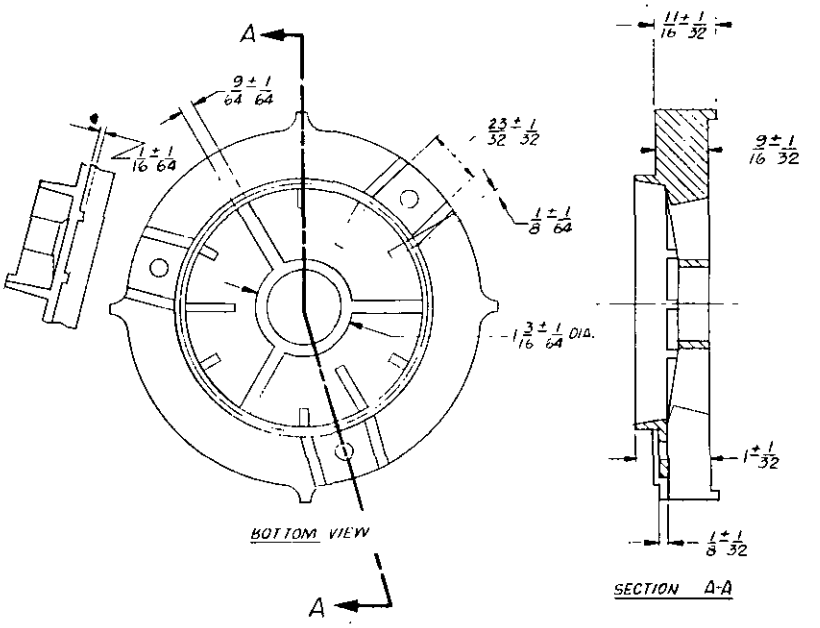
SOLDERING FRAME

FIG.-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:39Z
Check the source to verify that this is the current version before use.



SPIDER GRATE



SPIDER GRATE

FIG. 2

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:39Z
Check the source to verify that this is the currentversion before use.

3.2.2.2 Legs. The legs shall be of the design and dimensions shown in Figure 3. Legs may be furnished either angled or curved, as shown, to provide stability to the tripod. Holes for bolts shall be centered on the holes in the grate, and three bolts, 3/8 inch long, with nuts, shall be provided to facilitate rigid assembly of the tripod.

3.3 Finish.

3.3.1 Type I (Soldering frame). The wire shall have a polished or commercial, as drawn finish.

3.3.2 Type II (Tripod).

3.3.2.1 Spider grate. The spider grade shall be either cadmium plated or galvanized.

3.3.2.2 Legs. The legs shall be completely plated with either nickel, zinc or cadmium, and shall be smooth and bright.

3.3.2.3 Bolts and nuts. The bolts and nuts shall be plated with either nickel, zinc or cadmium and shall be smooth and bright.

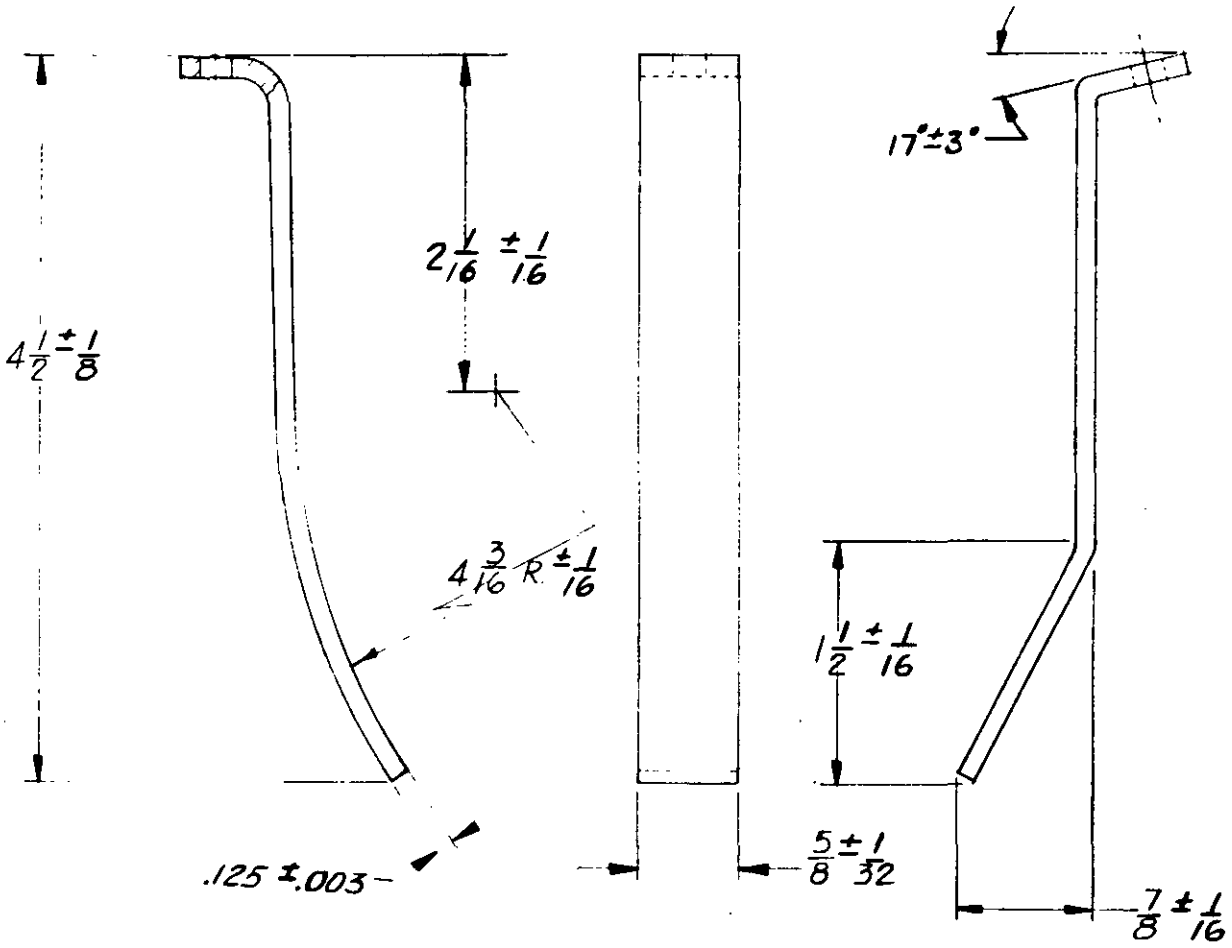
3.4 Distortion. The spider grate shall show no indications of distortion when subjected to the test described in 4.4.2. The distortion test is not required if lug or fracture test requirements of 3.1.2.1 have been satisfied.

3.5 Workmanship. Soldering frames shall be smoothly finished and free from sharp wire ends, or loose wire strands at edges. Tripods shall be symmetrical and stable. They shall be free from defects (cracks, cuts, scratches) or contamination which detract from their appearance or may impair their serviceability. The spider grate shall be free from significant casting defects such as injurious cold shuts, seams, flashes, blow holes or porosity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:39Z
Check the source to verify that this is the currentversion before use.



LEGS (3 REQUIRED)

ALTERNATE SHAPE
DIMS. ARE THE SAME EXCEPT AS NOTED.

NOTE:
$6\frac{11}{16} \pm \frac{1}{8}$ PERIPHERY OF LEGS WHEN ATTACHED TO SPIDER GRATE.

FIG.-3

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:39Z
Check the source to verify that this is the current version before use.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained,may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.1 For visual examination.

Unit of Product	Classification of defects	Inspection level	AQL % defective
Tripod and Soldering frame	Major	II	1.0
Tripod	Minor A	II	2.5
Tripod and Soldering frame	Minor B	S-2	2.5

4.2.2 For chemical analysis.

4.2.2.1 Soldering frame only. In the absence of a certificate of quality (4.1.2) sampling shall be in accordance with level S-1 MIL-STD-105, and Acceptance Number shall be zero. In all cases a single composite sample shall be obtained for analysis.

4.2.3 Distortion test.

4.2.3.1 Spider grate only. Sampling shall be in accordance with level S-2 and Acceptance Number shall be zero.

4.3 Examination. The tripods shall be examined to determine compliance with all requirements in Sections 3 and 5 of this specification. Nonconformance shall be permitted to the extent shown in 4.2.1.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:39Z
Check the source to verify that this is the current version before use.

4.3.1 Classification of defects.

Categories and defects:

Major

101 Tripod not stable.
102 Spider grate design for center collar ring and all radial ribs not according to Figure 2.
103 Spider grate working surface does not uniformly support a flat work load.
104 Spider grate shows injurious casting defects.
105 Legs not angled properly.
106 Legs not secured flatly or properly to spider grate by bolt and nut.
107 Tripod finish not metal specified.

Minor A

201 Tripod not symmetrical
202 Tripod finish not complete or uniform.
203 Soldering frame edge is not smooth.
204 Soldering frame outer wire strand is loose.
205 Spider grate not flat on level surface.
206 Bolts and nuts not as specified.
207 Soldering frame finish not as specified.

Minor B

251 Dimensions of soldering frame not as specified.
252 Wire size for soldering frame not as specified.
253 Soldering frame mesh not three wires per inch.
254 Dimensions of spider grate not as specified.
255 Dimensions of legs not as specified.

4.4 Test procedures.

4.4.1 Chemical analysis.

4.4.1.1 Soldering frame only. The composite sample, selected in accordance with 4.2.2.1 shall be analyzed in accordance with Methods 111 or 112 of Federal Test Method No. 151. In case of dispute, Method 111 shall be used for referee analysis.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:39Z
Check the source to verify that this is the currentversion before use.

4.4.2 Distortion.

4.4.2.1 Spider grate only. A spider grate which, after examination for flatness, complies with requirements shall be heated, one portion at a time, over a Bunsen burner or blow torch until the area reaches red heat, or a minimum temperature of 1100° F., and held at red heat for at least five minutes. Following this, the opposite portion of the grate shall be heated in the same manner, and this procedure repeated until all portions of the grate have been so heated. Upon cooling, the grate shall show no signs of distortion or evidence, by examination for flatness that the grate, after testing, fails to conform to the requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One soldering frame or one tripod, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Type	Unit package	Intermediate package	Exterior container
Type I-Frame	1 unit	12 units	144 units
Type II-Tripod	1 unit	Not required	36 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package.

5.1.3.1.1 Frame. Each frame shall be wrapped in neutral kraft paper or equal and packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors will be acceptable. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:39Z
Check the source to verify that this is the current version before use.

5.1.3.1.2 Tripod. Each tripod shall be disassembled and all disassembled components shall be preserved with VV-L-800 preservative. The preserved items shall be wrapped with MIL-B-121 barrier material and packaged in a carton of appropriate size conforming to PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. Carton shall contain sufficient cushioning material to prevent shifting of or damage to the contents. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:39Z
Check the source to verify that this is the current version before use.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Type required (see 1.2.1).
(c) Selection of applicable levels of packing. (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Category: N

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:39Z
Check the source to verify that this is the current version before use.